

第四章: 积分变换法

Fourier变换的来源

周期 T 的函数 f 可以展开成 Fourier 级数.

$$f(x) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} \left(A_n \sin \frac{2n\pi x}{T} + B_n \cos \frac{2n\pi x}{T} \right)$$

注: 若 f 定义在有限区间 $[a, b]$ 上, 可被认为是周期的(通过延拓), 周期为 $T=b-a$.

利用 Euler 公式:

$$\sin \frac{2n\pi x}{T} = \frac{1}{2i} \left(e^{\frac{2n\pi x}{T}i} - e^{-\frac{2n\pi x}{T}i} \right), \quad \cos \frac{2n\pi x}{T} = \frac{1}{2} \left(e^{\frac{2n\pi x}{T}i} + e^{-\frac{2n\pi x}{T}i} \right)$$

得:
$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{\frac{2n\pi x}{T}i}, \quad C_n \in \mathbb{C}$$

利用 $\{e^{\frac{2n\pi x}{T}i}\}$ 正交性:

$$\frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} e^{\frac{2\lambda(n-m)x}{T}i} dx = \delta_{nm}$$

可以算出

$$C_n = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(y) e^{-\frac{2n\pi y}{T}i} dy.$$

代入 f 的 Fourier 级数中:

$$\begin{aligned} f(x) &= \frac{1}{T} \sum_{n=-\infty}^{\infty} \left(\int_{-\frac{T}{2}}^{\frac{T}{2}} f(y) e^{-\frac{2n\pi y}{T}i} dy \right) e^{\frac{2n\pi x}{T}i} \\ &= \frac{1}{2\pi} \cdot \underbrace{\left(\frac{2\pi}{T} \right)}_{=\lambda_{n+1}-\lambda_n} \sum_{n=-\infty}^{\infty} \left(\int_{-\frac{T}{2}}^{\frac{T}{2}} f(y) e^{-\lambda_n y i} dy \right) e^{\lambda_n x i} \quad \left(\lambda_n = \frac{2\pi}{T}n \right) \end{aligned}$$

这是 $\lambda \mapsto e^{\lambda x i} F_T(\lambda)$ 关于分划 $\{\Delta_n\}_{n=0, \pm 1, \dots}$, $\Delta_n = [\lambda_n, \lambda_{n+1}]$ 的 Darboux 和.

其中:
$$F_T(\lambda) = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(y) e^{-\lambda y i} dy.$$

当 $T \rightarrow \infty$ 时,

$$F_T(\lambda) = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(y) e^{-\lambda y i} dy \rightarrow \hat{f}(\lambda) = \int_{\mathbb{R}} f(y) e^{-\lambda y i} dy.$$

$$\frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \rightarrow \int_{\mathbb{R}} d\lambda.$$

若 f 是 $\mathbb{R}$ 上的任何函数, 可以认为其周期 $T = \infty$. 在 Fourier 级数中

令 $T \rightarrow \infty$,

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\lambda) e^{\lambda x i} d\lambda$$

定义: Fourier 变换及其逆变换

若 $f \in L^1(\mathbb{R}^n)$ (即绝对可积),

其 Fourier 变换为 实变元的复值函数.

$$\mathcal{F}[f](\lambda) = \hat{f}(\lambda) = \int_{\mathbb{R}^n} f(x) e^{-\lambda x i} dx, \quad \forall \lambda \in \mathbb{R}^n$$

其 Fourier 逆变换为

$$\mathcal{F}^{-1}[f](x) = f^\vee(x) = \frac{1}{2\pi} \int_{\mathbb{R}^n} \hat{f}(\lambda) e^{\lambda x i} d\lambda, \quad \forall x \in \mathbb{R}^n$$

课程仅考虑 $n=1$ 的情况, 此时 $\lambda \cdot x = x\lambda$

Fourier 反演公式

若 $f \in L^1(\mathbb{R}) \cap C^1(\mathbb{R})$ (或分段 C^1)

则 $f = \mathcal{F}^{-1} \circ \mathcal{F}[f]$.

eg. 求 f 的 Fourier 变换: $f(x) = \begin{cases} 0 & , x \leq 0 \\ e^{-\beta x} (\beta > 0) & , x > 0 \end{cases}$

$$\begin{aligned}\hat{f}(\lambda) &= \int_{\mathbb{R}} f(x) e^{-i\lambda x} dx \\ &= \int_0^{\infty} e^{-(\beta + i\lambda)x} dx \\ &= -\frac{1}{\beta + i\lambda} e^{-(\beta + i\lambda)x} \Big|_0^{\infty} \\ &= \frac{1}{\beta + i\lambda}\end{aligned}$$

eg. $f(x) = e^{-|x|}$, 求其 Fourier 变换.

$$\begin{aligned}\hat{f}(\lambda) &= \int_{\mathbb{R}} f(x) e^{-i\lambda x} dx = \int_{\mathbb{R}} e^{-|x| - i\lambda x} dx \\ &= \int_{-\infty}^0 e^{(1-i\lambda)x} dx + \int_0^{\infty} e^{(-1-i\lambda)x} dx \\ &= \frac{1}{1-i\lambda} e^{(1-i\lambda)x} \Big|_{-\infty}^0 - \frac{1}{1+i\lambda} e^{(-1-i\lambda)x} \Big|_0^{\infty} \\ &= \frac{1}{1-i\lambda} + \frac{1}{1+i\lambda} = \frac{2}{1+\lambda^2}\end{aligned}$$

eg. $f(x) = \frac{\sin ax}{x}$ ($a > 0$), 求其 Fourier 变换.

$$\begin{aligned}\hat{f}(\lambda) &= \int_{\mathbb{R}} f(x) e^{-i\lambda x} dx = \int_{\mathbb{R}} \frac{\sin ax}{x} \cos(\lambda x) dx - i \int_{\mathbb{R}} \frac{\sin ax}{x} \sin(\lambda x) dx \\ &= 2 \int_0^{\infty} \frac{\sin ax}{x} \cos(\lambda x) dx \quad (\text{奇偶性}) \\ &= \int_0^{\infty} \frac{\sin(a+\lambda)x + \sin(a-\lambda)x}{x} dx \quad \text{注意到 } \int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2} \\ &= \pi, |\lambda| < a \quad \text{or} \quad = 0, |\lambda| > a \quad \text{or} \quad \frac{\pi}{2}, |\lambda| = a.\end{aligned}$$

我们验证

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

对任何 $(x, t) \in [0, \infty) \times [0, \infty)$, 考虑 $f(x, t) = \frac{\sin x}{x} e^{-tx}$, 以及 $F(t) = \int_0^{\infty} f(x, t) dt$ 。

$$F'(t) = - \int_0^{\infty} \sin x e^{-tx} dx.$$

利用分部积分,

$$\begin{aligned} \int_0^{\infty} \sin x e^{-tx} dx &= -[e^{-tx} \cos x]_0^{\infty} - t \int_0^{\infty} e^{-tx} \cos x dx \\ &= 1 - t \left([\sin x e^{-tx}]_0^{\infty} + t \int_0^{\infty} \sin x e^{-tx} dx \right) \\ &= 1 - t^2 \int_0^{\infty} \sin x e^{-tx} dx. \end{aligned}$$

因此得到 $F'(t) = -(1+t^2)^{-1} = -(\arctan t)'$ 。所以

$$\int_{[0, \infty)} \frac{\sin x}{x} dx = F(0) = F(\infty) + F(0) - F(\infty) = - \int_0^{\infty} F'(t) dt = \arctan x \Big|_0^{\infty} = \frac{\pi}{2}.$$

Fourier 变换的基本性质

记 $f^{\wedge} = \hat{f} = \mathcal{F}[f]$, $f^{\vee} = \mathcal{F}^{-1}[f]$.

1. 线性性 $(f + \mu g)^{\wedge} = \hat{f} + \mu \hat{g}$, $(f + \mu g)^{\vee} = f^{\vee} + \mu g^{\vee}$ ($\mu \in \mathbb{C}$)

2. 平移性 $[f(x-h)]^{\wedge}(\lambda) = e^{-i h \lambda} \hat{f}(\lambda)$, $= \int_{\mathbb{R}} f(x-h) e^{-i \lambda x} dx$
 $= \int_{\mathbb{R}} f(x-h) e^{-i \lambda(x-h)} \cdot e^{-i \lambda h} d(x-h)$

$$\begin{aligned} [f(x) e^{i h x}]^{\wedge}(\lambda) &= \hat{f}(\lambda-h) \quad (h \in \mathbb{R}) \\ &= \int_{\mathbb{R}} f(x) e^{i h x} \cdot e^{-i \lambda x} dx \\ &= \int_{\mathbb{R}} f(x) \cdot e^{-i(\lambda-h)x} dx \end{aligned}$$

3. 伸缩性 $[f(ax)]^{\wedge}(\lambda) = a^{-1} \hat{f}(a^{-1} \lambda)$, $a \in \mathbb{R} \setminus \{0\}$.

$$= \int_{\mathbb{R}} f(ax) \cdot e^{-i \lambda x} dx = \frac{1}{a} \cdot \int_{\mathbb{R}} f(x) \cdot e^{-i(\frac{\lambda}{a}) \cdot ax} d(ax) \quad (m \in \mathbb{N})$$

4. 乘子性质 $[x f(x)]^{\wedge}(\lambda) = i \frac{d}{d\lambda} \hat{f}(\lambda)$. 一般地, $[x^m f(x)]^{\wedge}(\lambda) = i^m \frac{d^m}{d\lambda^m} \hat{f}(\lambda)$

$$\begin{aligned} &= \int_{\mathbb{R}} x f(x) e^{-i \lambda x} dx = i \cdot \frac{d}{d\lambda} \int_{\mathbb{R}} f(x) e^{-i \lambda x} dx \\ &= i \cdot \int_{\mathbb{R}} (-ix) \cdot f(x) e^{-i \lambda x} dx \end{aligned}$$

5. 对称性质 $\hat{f}(-\lambda) = 2\pi f^{\vee}(\lambda)$.

6. 伴随性质: $\int_{\mathbb{R}} f(x) \hat{g}(x) dx = \int_{\mathbb{R}} \hat{f}(x) g(x) dx$

$$= \int_{\mathbb{R}} f(x) \left[\int_{\mathbb{R}} g(\lambda) e^{-i\lambda x} d\lambda \right] dx = \int_{\mathbb{R}} g(\lambda) \left[\int_{\mathbb{R}} f(x) e^{-i\lambda x} dx \right] d\lambda$$

7. 微分性质:

若 $f \in C^m(\mathbb{R})$, $\lim_{|x| \rightarrow \infty} f^{(k)}(x) = 0$, $(k=1, \dots, m-1)$.

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$$[f^{(m)}]^{\wedge}(\lambda) = (i\lambda)^m f^{\wedge}(\lambda).$$

导数 $\rightarrow$ 多项式.

$$= \int_{\mathbb{R}} f^{(m)} e^{-i\lambda x} dx = \int_{\mathbb{R}} e^{-i\lambda x} d f^{(m-1)} = i\lambda \int_{\mathbb{R}} f^{(m-1)} e^{-i\lambda x} dx = \dots = (i\lambda)^m f^{\wedge}(\lambda).$$

8. 积分性质:

若 $g(x) = \int_{-\infty}^x f(t) dt \in L'(\mathbb{R})$, 则 $\hat{g}(\lambda) = \frac{1}{i\lambda} \hat{f}(\lambda)$.

$$f = g', \quad \hat{f}(\lambda) = [g']^{\wedge}(\lambda) = i\lambda \hat{g}(\lambda).$$

9. Parseval 等式:

若 $f, \hat{f} \in L'$, 则 $|\hat{f}|^{\vee} = f$, 则 $\int_{\mathbb{R}} |f|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}|^2$.

$$= \int_{\mathbb{R}} f(x) \left[\frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\lambda) e^{i\lambda x} d\lambda \right] dx$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\lambda) \left[\int_{\mathbb{R}} f(x) e^{i\lambda x} dx \right] d\lambda$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}[\hat{f}] \cdot \overline{\mathcal{F}[\hat{f}]}$$

卷积定义及其运算

def $f, g \in L^1(\mathbb{R}^n)$, $\Rightarrow$ 卷积 (convolution) 为

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x-y) g(y) dy$$

$$= \int_{\mathbb{R}^n} f(x-y) g(y) dy = \int_{\mathbb{R}^n} f(x+y) g(-y) dy = \int_{\mathbb{R}^n} f(z) g(x-z) dz$$

注1: 卷积是对称的, $(f * g)(x) = (g * f)(x) = (g * f)(x)$

注2: 课程仅考虑 $n=1$, $f * g(x) = \int_{\mathbb{R}} f(x-y) g(y) dy$

eg. $f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$; $g(x) = \begin{cases} 0, & x < 0 \\ e^{-x}, & x \geq 0 \end{cases}$.

$$\begin{aligned} (f * g)(x) &= \int_{\mathbb{R}} f(x-y)g(y)dy \\ &= \int_{\mathbb{R}} 1_{\{y \leq x\}} \cdot 1_{\{y \geq 0\}} e^{-y} dy \\ &= \int_{\mathbb{R}} 1_{\{0 \leq y \leq x\}} e^{-y} dy \\ &= \begin{cases} 0, & x \leq 0 \\ \int_0^x e^{-y} dy, & x > 0 \end{cases} = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x}, & x > 0. \end{cases} \end{aligned}$$

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Fourier 变换的卷积性质

若 $f, g \in L^1$, 则 $\mathcal{F}[f * g] = \mathcal{F}[f] \mathcal{F}[g]$.

$$\begin{aligned} & \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(x-y)g(y)dy \right] e^{-i\lambda x} dx \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(x-y) e^{-i\lambda x} dx \right] g(y) dy \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(x-y) e^{-i\lambda(x-y)} dx \right] e^{-i\lambda y} g(y) dy \\ &= \int_{\mathbb{R}} \hat{f}(\lambda) e^{-i\lambda y} g(y) dy = \hat{f}(\lambda) \hat{g}(\lambda) \end{aligned}$$

$\mathcal{F}^{-1}\{\mathcal{F}[f * g]\} = \mathcal{F}^{-1}\{\mathcal{F}[f] \mathcal{F}[g]\}$
 $f * g = \mathcal{F}^{-1}\{\mathcal{F}[f] \cdot \mathcal{F}[g]\}$ 推论:
 $\mathcal{F}[f * g] = \mathcal{F}[f] \mathcal{F}[g]$

$\mathcal{F}^{-1}[\mathcal{F}[f] \mathcal{F}[g]] = \mathcal{F}^{-1}[\mathcal{F}[f] * \mathcal{F}[g]]$.

eg. 求 $f(x) = e^{-x^2}$ 的 Fourier 变换.

$$\begin{aligned} \hat{f}(\lambda) &= \int_{\mathbb{R}} e^{-x^2} e^{-i\lambda x} dx = \frac{i}{\lambda} \int_{\mathbb{R}} e^{-x^2} d e^{-i\lambda x} \\ &= \frac{i}{\lambda} [e^{-x^2} e^{-i\lambda x}] \Big|_{x=-\infty}^{x=\infty} + \frac{2i}{\lambda} \int_{\mathbb{R}} x e^{-x^2} e^{-i\lambda x} dx \\ &= \frac{2i}{\lambda} \int_{\mathbb{R}} x e^{-x^2} e^{-i\lambda x} dx = \frac{2i}{\lambda} [x f(x)]^{\wedge}(\lambda) \\ &= \frac{2i}{\lambda} \cdot i \cdot \frac{d}{d\lambda} \hat{f}(\lambda) \end{aligned}$$

$\Rightarrow \hat{f}(\lambda)$ 满足 $\frac{d}{d\lambda} \hat{f}(\lambda) = -\frac{\lambda}{2} \hat{f}(\lambda) \Rightarrow \hat{f}(\lambda) = \hat{f}(0) e^{-\frac{1}{4}\lambda^2} = \sqrt{\pi} e^{-\frac{1}{4}\lambda^2}$.

其中 $\hat{f}(0) = \int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\int_{\mathbb{R}^2} e^{-x^2-y^2} dx dy} = \sqrt{2\pi \int_0^{\infty} e^{-r^2} r dr} = \sqrt{\pi \int_0^{\infty} e^{-t} dt} = \sqrt{\pi}$.

ex. 求 $f(x) = e^{-3(x^2+4x-2)}$ 的 Fourier 变换.

$$\begin{aligned}\hat{f}(\lambda) &= \int_{\mathbb{R}} e^{-3(x+2)^2+18} \cdot e^{-i\lambda x} dx \\&= e^{18} \cdot \int_{\mathbb{R}} e^{-3(x+2)^2} \cdot e^{-i\lambda x} dx \\&= e^{18} \cdot \int_{\mathbb{R}} e^{-[\sqrt{3}(x+2)]^2} \cdot e^{-i[\sqrt{3}(x+2)] \frac{\lambda}{\sqrt{3}}} \cdot e^{2i\lambda} dx \\&= e^{18+2i\lambda} \int_{\mathbb{R}} e^{-z^2} \cdot e^{-iz \cdot \frac{\lambda}{\sqrt{3}}} d\left(\frac{z}{\sqrt{3}}\right) \\&= \frac{e^{18+2i\lambda}}{\sqrt{3}} \int_{\mathbb{R}} e^{-x^2} \cdot e^{-ix \cdot (\frac{\lambda}{\sqrt{3}})} dx \\&= \frac{e^{18+2i\lambda}}{\sqrt{3}} \hat{g}\left(\frac{\lambda}{\sqrt{3}}\right), \quad \hat{g}(\lambda) = \mathcal{F}[e^{-x^2}] = \sqrt{\pi} e^{-\frac{\lambda^2}{4}} \\&= \frac{e^{18+2i\lambda}}{\sqrt{3}} \cdot \sqrt{\pi} e^{-\frac{\lambda^2}{12}} = \sqrt{\frac{\pi}{3}} e^{18 - \frac{\lambda^2}{12} + 2i\lambda}.\end{aligned}$$

Fourier 变换在求解 PDE 初值问题中的应用

core: $[f^{(n)}]^\wedge(\lambda) = (i\lambda)^n \hat{f}(\lambda)$. PDE $\rightarrow$ ODE .

I: 一维热传导方程的柯西问题

$$\begin{cases} \partial_t u - a^2 \partial_x^2 u = f(x, t), & x \in \mathbb{R}, t > 0 \\ u(x, 0) = \phi(x), & x \in \mathbb{R} \end{cases}$$

关于 x 做 Fourier 变换, $\hat{u}(\lambda, t) = \mathcal{F}_x[u(x, t)](\lambda)$

$$\begin{aligned}(\text{把 } t \text{ 视为参数}) \quad \mathcal{F}_x[\partial_t u(x, t)](\lambda) &= \int_{\mathbb{R}} \partial_t u(x, t) e^{-i\lambda x} dx \\&= \partial_t \left[\int_{\mathbb{R}} u(x, t) e^{-i\lambda x} dx \right] = \frac{d}{dt} \hat{u}(\lambda, t).\end{aligned}$$

$$\mathcal{F}_x[\partial_x^2 u(x, t)](\lambda) = (i\lambda)^2 \hat{u}(\lambda, t) = -\lambda^2 \hat{u}(\lambda, t)$$

$$\mathcal{F}_x[f(x, t)](\lambda) = \hat{f}(\lambda, t).$$

把 $u(x, t)$ 的 PDE $\rightarrow \hat{u}(\lambda, \cdot)$ 关于 t 的 ODE (λ 视为参数)

$$(\star) \begin{cases} \frac{d}{dt} \hat{u} + a^2 \lambda^2 \hat{u} = \hat{f} & , t > 0 \\ \hat{u}|_{t=0} = \hat{\phi}(\lambda) \end{cases}$$

由此得 $\hat{u}(\lambda, t) = \hat{\phi}(\lambda) e^{-a^2 \lambda^2 t} + \int_0^t \hat{f}(\lambda, \tau) e^{-a^2 \lambda^2 (t-\tau)} d\tau$

$$\begin{aligned} \text{则 } u(x, t) &= \mathcal{F}_\lambda^{-1} [\hat{u}(\lambda, t)](x) \\ &= \mathcal{F}_\lambda^{-1} [\hat{\phi}(\lambda) e^{-a^2 \lambda^2 t}] + \int_0^t \mathcal{F}_\lambda^{-1} [\hat{f}(\lambda, \tau) e^{-a^2 \lambda^2 (t-\tau)}] d\tau \end{aligned}$$

$$[e^{-x^2}]^\wedge(\lambda) = \sqrt{\pi} e^{-\frac{\lambda^2}{4}} \Rightarrow \frac{1}{2\sqrt{\pi}} \int_{\mathbb{R}} e^{-\frac{\lambda^2}{4}} e^{i\lambda x} d\lambda = e^{-x^2}$$

$$\begin{aligned} \text{因此, } \mathcal{F}_\lambda^{-1} [e^{-a^2 \lambda^2 t}](x) &= \frac{1}{2\pi} \int_{\mathbb{R}} e^{-a^2 \lambda^2 t} e^{-i\lambda x} d\lambda \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} e^{-\frac{(2a\sqrt{t}\lambda)^2}{4}} e^{i(2a\sqrt{t}\lambda) \frac{x}{2a\sqrt{t}}} d\lambda \\ &= \frac{1}{2a\sqrt{\pi t}} e^{-\frac{x^2}{4a^2 t}} \quad \text{热方程的基本解} \end{aligned}$$

继续计算 $\hat{u}$ 的 Fourier 变换. $(fg)^\vee = f^\vee * g^\vee$

$$\begin{aligned} \mathcal{F}_\lambda^{-1} [\hat{\phi}(\lambda) e^{-a^2 \lambda^2 t}](x) &= \mathcal{F}_\lambda^{-1} [\hat{\phi}] * \mathcal{F}_\lambda^{-1} [e^{-a^2 \lambda^2 t}] \\ &= \frac{1}{2a\sqrt{\pi t}} \int_{\mathbb{R}} \phi(y) e^{-\frac{(x-y)^2}{4a^2 t}} dy \end{aligned}$$

$$\begin{aligned} \mathcal{F}_\lambda^{-1} [\hat{f}(\lambda, \tau) e^{-a^2 \lambda^2 (t-\tau)}](x) &= \mathcal{F}_\lambda^{-1} [\hat{f}(\lambda, \tau)] * \mathcal{F}_\lambda^{-1} [e^{-a^2 \lambda^2 (t-\tau)}] \\ &= \frac{1}{2a\sqrt{\pi(t-\tau)}} \int_{\mathbb{R}} f(y, \tau) e^{-\frac{(x-y)^2}{4a^2(t-\tau)}} dy \end{aligned}$$

则热方程的柯西问题的解为

$$u(x, t) = \frac{1}{2a\sqrt{\pi t}} \int_{\mathbb{R}} \phi(y) e^{-\frac{(x-y)^2}{4a^2 t}} dy + \frac{1}{2a\sqrt{\pi}} \int_0^t \frac{1}{\sqrt{t-\tau}} \int_{\mathbb{R}} f(y, \tau) e^{-\frac{(x-y)^2}{4a^2(t-\tau)}} dy d\tau$$

注: 1. 可以推广至一般维数 n ; 2. 可用反射原理解半直线上的热方程;

3. 该公式说明解在任何点的值瞬时受到初值影响, 与波方程不同.

II: 上半平面上的调和函数.

特征线法: 波方程.

分离变量法: 有界域上

$$\begin{cases} \Delta u(x, y) = 0, & (x, y) \in \mathbb{R} \times \mathbb{R}_+ \\ u(x, 0) = f(x), & x \in \mathbb{R} \\ |u(x, y)| \text{ 有界}, & (x, y) \in \mathbb{R} \times \mathbb{R}_+ \end{cases}$$

谁定义在有界域上就关于谁做.

关于 x 做 Fourier 变换, $\hat{u}(\lambda, y) = \mathcal{F}_x[u(x, y)](\lambda)$. (y 视为参数).

$$\mathcal{F}_x[\partial_x^2 u(x, y)](\lambda) = (-\lambda)^2 \hat{u}(\lambda, y)$$

$$\mathcal{F}_x[\partial_y^2 u(x, y)](\lambda) = \frac{d^2}{dy^2} \hat{u}(\lambda, y).$$

$$\mathcal{F}_x[u(x, 0)](\lambda) = \hat{f}(\lambda).$$

$$\Rightarrow \begin{cases} \frac{d^2}{dy^2} \hat{u} - \lambda^2 \hat{u} = 0, & y > 0 \\ \hat{u}|_{y=0} = \hat{f}(\lambda) \end{cases}, \quad \hat{u}(\lambda, y) \stackrel{\text{通解}}{=} A e^{-\lambda y} + B e^{\lambda y}$$

$\stackrel{\text{有界性}}{\stackrel{\text{初值条件}}{=}} \hat{f}(\lambda) e^{-|\lambda| y}$

为求 $\hat{u}$ 的 Fourier 逆变换, 首先计算:

$$\begin{aligned} 2\pi \mathcal{F}_\lambda^{-1}[e^{-|\lambda| y}](x) &= \int_{\mathbb{R}} e^{-|\lambda| y + i\lambda x} d\lambda \\ &= \int_{-\infty}^0 e^{(y+i\lambda)x} d\lambda + \int_0^{\infty} e^{(-y+i\lambda)x} d\lambda \\ &= \frac{1}{y+i\lambda} e^{(y+i\lambda)x} \Big|_{\lambda=-\infty}^{\lambda=0} + \frac{1}{-y+i\lambda} e^{(-y+i\lambda)x} \Big|_{\lambda=0}^{\lambda=\infty} \\ &= \frac{1}{y+i\lambda} + \frac{1}{y-i\lambda} = \frac{2y}{x^2+y^2}. \end{aligned}$$

该函数称为上半平面上 Laplace 方程的 Green 函数.

$$\begin{aligned} \mathcal{F}_\lambda^{-1}[\hat{f}(\lambda) e^{-|\lambda| y}](x) &= \mathcal{F}_\lambda^{-1}[\hat{f}(\lambda)] * \mathcal{F}_\lambda^{-1}[e^{-|\lambda| y}](x) \\ &= \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(z) y}{(x-z)^2 + y^2} dz. \end{aligned}$$

Poisson 公式:

$$u(x, y) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(z) y}{(x-z)^2 + y^2} dz.$$

(可推广至 $\mathbb{R}^{n-1} \times \mathbb{R}_+$)

Fourier 变换解偏微 (III): 一维波动方程的柯西问题

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u, & x \in \mathbb{R}, t > 0, \\ u(x, 0) = \phi(x), \partial_t u(x, 0) = \psi(x), & x \in \mathbb{R}. \end{cases}$$

关于 x 做 Fourier 变换, 记 $\hat{u}(\lambda, t) = \mathcal{F}_x[u(x, t)](\lambda)$, 用如下事实 (把 t 视为参数!!)

$$\mathcal{F}_x[\partial_t^2 u(x, t)](\lambda) = \frac{d^2}{dt^2} \hat{u}(\lambda, t),$$

$$\mathcal{F}_x[\partial_x^2 u(x, t)](\lambda) = (i\lambda)^2 \hat{u}(\lambda, t), \text{ 微分性质.}$$

$$\mathcal{F}_x[u(x, 0)](\lambda) = \hat{\phi}(\lambda), \quad \mathcal{F}_x[\partial_t u(x, 0)](\lambda) = \frac{d}{dt} \hat{u}(\lambda, 0) = \hat{\psi}(\lambda),$$

我们把 $u(x, t)$ 的偏微转化为 $\hat{u}(\lambda, \cdot)$ 关于 t 的如下常微 (把 λ 视为参数!!)

$$(\star) \begin{cases} \frac{d^2}{dt^2} \hat{u} + a^2 \lambda^2 \hat{u} = 0, & t > 0, \\ \hat{u}|_{t=0} = \hat{\phi}(\lambda), \frac{d}{dt} \hat{u}|_{t=0} = \hat{\psi}(\lambda). \end{cases}$$

容易得到(★)的通解为 $\hat{u}(\lambda, t) = A \cos a\lambda t + B \sin a\lambda t$ 。利用常微初值,

$$\hat{u}(\lambda, t) = \hat{\phi}(\lambda) \cos a\lambda t + \frac{\hat{\psi}(\lambda)}{a\lambda} \sin a\lambda t.$$

为求 $\hat{u}$ 的 Fourier 逆变换, 我们计算

$$\begin{aligned} \mathcal{F}_\lambda^{-1}[\hat{\phi}(\lambda) \cos a\lambda t](x) &= \frac{1}{4\pi} \int_{\mathbb{R}} \hat{\phi}(\lambda) [e^{a\lambda ti} + e^{-a\lambda ti}] e^{i\lambda x} d\lambda \\ &= \frac{1}{4\pi} \int_{\mathbb{R}} \hat{\phi}(\lambda) e^{i\lambda(x+at)} d\lambda + \frac{1}{4\pi} \int_{\mathbb{R}} \hat{\phi}(\lambda) e^{i\lambda(x-at)} d\lambda \\ &= \frac{1}{2} \left(\phi(x+at) + \phi(x-at) \right). \end{aligned}$$

Fourier 逆变换公式.

另一方面, 类似可得

$$\begin{aligned} \mathcal{F}_\lambda^{-1}\left[\frac{\hat{\psi}(\lambda)}{a\lambda} \sin a\lambda t\right](x) &= \frac{1}{2\pi} \int_{\mathbb{R}} \frac{\hat{\psi}(\lambda)}{2a\lambda i} [e^{a\lambda ti} - e^{-a\lambda ti}] e^{i\lambda x} d\lambda \\ &= \frac{1}{4a\pi} \int_{\mathbb{R}} \frac{\hat{\psi}(\lambda)}{\lambda i} \int_{x-at}^{x+at} \frac{d}{d\tau} e^{i\lambda \tau} d\tau d\lambda \\ &= \frac{1}{4a\pi} \int_{x-at}^{x+at} \int_{\mathbb{R}} \hat{\psi}(\lambda) e^{i\lambda \tau} d\lambda d\tau = \frac{1}{2a} \int_{x-at}^{x+at} \psi(\tau) d\tau. \end{aligned}$$

$= e^{i\lambda(x+at)} - e^{i\lambda(x-at)} = e^{i\lambda \tau} \Big|_{\tau=x-at}^{\tau=x+at} = \int_{x-at}^{x+at} (i\lambda) e^{i\lambda \tau} d\tau$

合并上述结果, 我们再次得到 d'Alembert 公式

$$u(x, t) = \frac{1}{2} \left(\phi(x+at) + \phi(x-at) \right) + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\tau) d\tau.$$

(2011年)

ex. 已知 $\mathcal{F}[e^{-x^2}](\lambda) = \sqrt{\pi} e^{-\frac{\lambda^2}{4}}$, $\int_{\mathbb{R}} f^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}|^2$.

假设 $\phi, \psi \in C_0^\infty(\mathbb{R})$. 证明:

$$\begin{cases} \partial_t^2 u - \partial_t^4 u = 0, & x \in \mathbb{R}, t > 0. \\ u(x, 0) = \phi(x), \partial_t u(x, 0) = \psi(x), & x \in \mathbb{R}. \end{cases}$$

$u(x, 0) = \phi(x), \partial_t u(x, 0) = \psi(x), x \in \mathbb{R}.$

Parseval 等式

存在有限能量解 u (即 $\sup_{t \geq 0} \int_{\mathbb{R}} |u(x, t)|^2 dx < \infty$)

的必要条件是: $\psi(x) = \phi_{xx}(x)$

并求出该有限能量解 $u(x, t)$.

关于 x 做 Fourier 变换, $\mathcal{F}_x[\partial_t^2 u(x, t)](\lambda) = \partial_t^2 \hat{u}(\lambda, t)$.

$$\mathcal{F}_x[\partial_t^4 u(x, t)](\lambda) = (-\lambda)^4 \hat{u}(\lambda, t).$$

$$\mathcal{F}_x[u(x, 0)](\lambda) = \hat{\phi}(\lambda), \mathcal{F}_x[\partial_t u(x, 0)] = \frac{d}{dt} \hat{u}(\lambda, 0) = \hat{\psi}(\lambda)$$

$$\text{例 } \begin{cases} \frac{d^2}{dt^2} \hat{u} - \lambda^4 \hat{u} = 0 \\ \hat{u}|_{t=0} = \hat{\phi}(\lambda), \frac{d}{dt} \hat{u}|_{t=0} = \hat{\psi}(\lambda). \end{cases}$$

$$\text{得 } \hat{u} = \frac{\hat{\phi}(\lambda) + \frac{\hat{\psi}(\lambda)}{\lambda^2}}{2} e^{\lambda^2 t} + \frac{\hat{\phi}(\lambda) - \frac{\hat{\psi}(\lambda)}{\lambda^2}}{2} e^{-\lambda^2 t}.$$

$\downarrow A(\lambda) \qquad \qquad \qquad \downarrow B(\lambda)$

1° 必要性: 由 Parseval 等式

$$|z_1 + z_2|^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = |z_1|^2 + |z_2|^2 + z_1 \bar{z}_2 + z_2 \bar{z}_1$$

$$\begin{aligned} C &\geq 2\pi \int_{\mathbb{R}} |u(x, t)|^2 dx = \int_{\mathbb{R}} |\hat{u}(\lambda, t)|^2 d\lambda = \int_{\mathbb{R}} |A|^2 e^{2\lambda^2 t} d\lambda + \int_{\mathbb{R}} |B|^2 e^{-2\lambda^2 t} d\lambda + \int_{\mathbb{R}} (A\bar{B} + \bar{A}B) d\lambda \\ &\geq \int_{\mathbb{R}} |A|^2 e^{2\lambda^2 t} d\lambda - C' \\ &\geq e^{2\epsilon^2 t} \int_{\mathbb{R}} |A|^2 d\lambda - C' \end{aligned}$$

t 越大, $e^{-2\lambda^2 t}$ 越小, 有界. $A\bar{B} + \bar{A}B$ 有界

由于 $e^{2\epsilon^2 t}$ 关于 t 无界, 且 $\epsilon > 0$, 例 $|A|^2 = 0, A(\lambda) = 0 = \hat{\phi}(\lambda) + \frac{\hat{\psi}(\lambda)}{\lambda^2} = \frac{1}{\lambda^2} (\lambda^2 \hat{\phi}(\lambda) + \hat{\psi}(\lambda)) = \frac{1}{\lambda^2} \mathcal{F}[\psi - \phi_{xx}]$

由 Fourier 逆变换, 存在有限能量解的必要条件是

$$\psi = \phi_{xx}.$$

2° 充分性.

若 $\psi(x) = \phi_{xx}(x)$, 例 $\hat{\psi}(\lambda) = (-\lambda)^2 \hat{\phi}(\lambda) = -\lambda^2 \hat{\phi}(\lambda)$, $\hat{\phi}(\lambda) + \frac{\hat{\psi}(\lambda)}{\lambda^2} = 0$.

$\hat{u} = \hat{\phi}(\lambda) e^{-\lambda^2 t}$. 由 Parseval 等式:

$$\int_{\mathbb{R}} |u(x,t)|^2 dx = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{\phi}(\lambda)|^2 e^{-2\lambda^2 t} d\lambda \leq \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{\phi}(\lambda)|^2 d\lambda = \int_{\mathbb{R}} \phi^2(x) dx < \infty. \quad \#$$

(已知 ϕ 在有界域上非零)

下面计算 $\hat{u}$ 的 Fourier 逆变换.

$$\begin{aligned} \mathcal{F}^{-1}[e^{-\lambda^2 t}](x) &= \frac{1}{2\pi} \int_{\mathbb{R}} e^{-t \cdot \lambda^2} \cdot e^{i x \lambda} d\lambda \\ &= \frac{1}{2\sqrt{\lambda t}} \cdot \frac{1}{\sqrt{\lambda}} \int_{\mathbb{R}} \sqrt{\lambda} \cdot e^{-\frac{1}{4} \cdot (2\sqrt{\lambda} \lambda)^2} \cdot e^{i \left(\frac{x}{2\sqrt{\lambda}}\right) (2\sqrt{\lambda} \lambda)} d(2\sqrt{\lambda} \lambda) \\ &= \frac{1}{2\sqrt{\lambda t}} \cdot e^{-\frac{x^2}{4t}}. \end{aligned}$$

由卷积性质: $u(x,t) = \frac{1}{2\sqrt{\lambda t}} \int_{\mathbb{R}} \phi(y) e^{-\frac{(x-y)^2}{4t}} dy.$

(2022秋)
ex. $\begin{cases} u_t - u_{xx} + x u_x = 0 & , (x,t) \in \mathbb{R} \times [0, \infty) \\ u|_{t=0} = \phi(x) & , x \in \mathbb{R}. \end{cases}$

关于 x 做 Fourier 变换, $\mathcal{F}_x[u_t(x,t)](\lambda) = \hat{u}_t(\lambda, t).$

$$\mathcal{F}_x[u_{xx}(x,t)](\lambda) = (i\lambda)^2 \hat{u}(\lambda, t) = -\lambda^2 \hat{u}(\lambda, t)$$

$$\mathcal{F}_x[x u_x](\lambda) = i \cdot \frac{\partial}{\partial \lambda} [i\lambda \cdot \hat{u}(\lambda, t)] = -\frac{\partial(\lambda \hat{u}(\lambda, t))}{\partial \lambda}.$$

$$\begin{cases} \frac{\partial \hat{u}}{\partial t} + \lambda^2 \hat{u} - \frac{\partial(\lambda \hat{u})}{\partial \lambda} = 0 \\ \hat{u}|_{t=0} = \hat{\phi}(\lambda) \end{cases}$$

取 $v(\lambda, t) = \lambda \hat{u}(\lambda, t).$

$$\begin{cases} \frac{1}{\lambda} \cdot \frac{\partial v}{\partial t} + \lambda \cdot v - \frac{\partial v}{\partial \lambda} = 0 & \Rightarrow \partial_t v - \lambda \partial_\lambda v + \lambda v = 0. \\ v|_{t=0} = \lambda \hat{\phi}(\lambda) \end{cases}$$

找一条线, PDE 在该线上是 ODE.

例 - 解 PDE (特征线法): $\frac{d\lambda}{dt} = -\lambda, \lambda = \lambda_0 e^{-t},$ 考虑 $H(t) = v(\lambda_0 e^{-t}, t).$

$$\frac{d}{dt} H(t) = \partial_\lambda v \cdot \lambda_0 \cdot (-e^{-t}) + \partial_t v = -\lambda \partial_\lambda v + \partial_t v = -\lambda^2 v = -\lambda_0^2 e^{-2t} \cdot H(t)$$

$$H'(t) + \lambda_0^2 e^{-2t} H(t) = 0$$

$$H(x) = C \cdot e^{-\int_0^x \lambda_0^2 e^{-2t} dt} = C \cdot e^{\frac{\lambda_0^2 (e^{-2x} - 1)}{2}}$$

$$H(0) = \lambda_0 \hat{\varphi}(\lambda_0) = C, \quad \lambda_0 = \lambda e^t$$

$$\text{则 } v(\lambda, t) = \lambda e^t \cdot \hat{\varphi}(\lambda e^t) e^{-\frac{1}{2} \lambda^2 (e^{2t} - 1)}$$

$$\hat{u}(\lambda, t) = e^t \cdot \hat{\varphi}(\lambda e^t) \cdot e^{-\frac{1}{2} (e^{2t} - 1) \lambda^2}$$

$$\begin{aligned} \mathcal{F}^{-1}[\hat{\varphi}(\lambda e^t)](x) &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{\varphi}(\lambda e^t) e^{ix\lambda} d\lambda = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{\varphi}(s) e^{ix s e^{-t}} d(s e^{-t}) \\ &= \frac{e^{-t}}{2\pi} \int_{\mathbb{R}} \hat{\varphi}(s) e^{i(x e^{-t}) \cdot s} ds = e^{-t} \varphi(x e^{-t}) \end{aligned}$$

另: 伸缩性质 $\mathcal{F}[f(x e^{-t})](\lambda) = e^t \cdot \hat{f}(\lambda \cdot e^t)$

$$\begin{aligned} \mathcal{F}^{-1}[e^{-\frac{1}{2} (e^{2t} - 1) \lambda^2}] &= \frac{1}{2\pi} \int_{\mathbb{R}} e^{-\frac{1}{2} (e^{2t} - 1) \lambda^2} \cdot e^{ix\lambda} d\lambda \\ &= \frac{1}{\sqrt{2\pi(e^{2t} - 1)}} \left\{ \frac{1}{2\pi} \int_{\mathbb{R}} \sqrt{\lambda} e^{-\frac{1}{4} (\sqrt{2(e^{2t} - 1)} \lambda)^2} \cdot e^{i \frac{x}{\sqrt{2(e^{2t} - 1)}} \cdot \sqrt{2(e^{2t} - 1)} \lambda} d(\sqrt{2(e^{2t} - 1)} \lambda) \right\} \\ &= \frac{1}{\sqrt{2\pi(e^{2t} - 1)}} e^{-\frac{x^2}{2(e^{2t} - 1)}} \end{aligned}$$

$$\begin{aligned} \text{由卷积性质, } u(x, t) &= e^t \cdot \int_{\mathbb{R}} e^{-t} \varphi(y e^{-t}) \cdot \frac{1}{\sqrt{2\pi(e^{2t} - 1)}} e^{-\frac{(x-y)^2}{2\pi(e^{2t} - 1)}} dy \\ &= \frac{1}{\sqrt{2\pi(e^{2t} - 1)}} \int_{\mathbb{R}} \varphi(y e^{-t}) e^{-\frac{(x-y)^2}{2\pi(e^{2t} - 1)}} dy \end{aligned}$$

Fourier 变换要求函数在全空间上有定义。若研究半无界问题，我们引入 Laplace 变换。

Laplace 变换

若 $f(t)$ 定义在 $\mathbb{R}_+ = [0, \infty)$ 上。任取充分大的 $r > 0$ ，考虑函数 $f_r: \mathbb{R} \rightarrow \mathbb{R}$,

$$f_r(t) = \begin{cases} e^{-rt}f(t), & t \geq 0 \\ 0, & t < 0. \end{cases}$$

对 $f_r(t)$ 做 Fourier 变换

$$\widehat{f_r}(\lambda) = \int_{\mathbb{R}} f_r(t) e^{-i\lambda t} dt = \int_0^{\infty} f(t) e^{-(r+i\lambda)t} dt = \int_0^{\infty} f(t) e^{-st} dt,$$

其中 $s = r + i\lambda$ 。定义上式的右端项为：

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt.$$

由 Fourier 逆变换，并利用 $ds = i d\lambda$ ，我们得到

$$\begin{aligned} f_r(t) &= [\widehat{f_r}]^{\vee}(t) = \frac{1}{2\pi} \int_{\mathbb{R}} F(r + i\lambda) e^{i\lambda t} d\lambda = \frac{1}{2\pi i} \int_{\{r+i\mathbb{R}\}} F(s) e^{(s-r)t} ds \\ \Rightarrow f(t) &= f_r(t) e^{rt} = \frac{1}{2\pi i} \int_{\{r+i\mathbb{R}\}} F(s) e^{st} ds. \end{aligned}$$

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定义：Laplace 变换及其逆变换

- 若 $f \in L^1_{\text{loc}}(\mathbb{R}_+)$ ，其 Laplace 变换为

$$\mathcal{L}[f](s) = F(s) = \int_0^{\infty} f(t) e^{-st} dt, \quad s \in \mathbb{C}.$$

注：我们仅考虑使得该积分有定义的 $s \in \mathbb{C}$ 。

- 若 $F(s)$ 是函数 $f(t)$ 的 Laplace 变换，则称 $f(t)$ 为 $F(s)$ 的 Laplace 逆变换，记为

$$f(t) = \mathcal{L}^{-1}[F(s)](t).$$

存在性定理

- 若 f 在 $\mathbb{R}_+$ 上分段连续，且 $|f(t)| \leq M e^{s_0 t} (\forall t \in \mathbb{R}_+)$ ，则

$$\mathcal{L}[f](s) = F(s) = \int_0^{\infty} f(t) e^{-st} dt \quad \text{在半平面 } \operatorname{Re} s > s_0 \text{ 上存在.}$$

- 假设 f 同上述条件， $F(s) = \mathcal{L}[f](s)$ 。如果 $t > 0$ 为 f 的连续点，则

$$f(t) = \frac{1}{2\pi i} \int_{\{r+i\mathbb{R}\}} F(s) e^{st} ds, \quad \forall r > s_0.$$

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Laplace 变换的基本性质

假设下面涉及的所有函数均满足 $Me^{s_0 t}$ 的增长控制；且其 Laplace 变换和逆变换均存在。

- 线性性： $\mathcal{L}[f + \lambda g] = \mathcal{L}[f] + \lambda \mathcal{L}[g]$, $\mathcal{L}^{-1}[f + \lambda g] = \mathcal{L}^{-1}[f] + \lambda \mathcal{L}^{-1}[g]$ ($\lambda \in \mathbb{C}$)。
- 平移性： 对任何 $h > 0$, $\mathcal{L}[f(t-h)](s) = e^{-hs} \mathcal{L}[f](s)$ 。
- 位移性质： 对任何 $a \in \mathbb{R}$, $\mathcal{L}[e^{at}f(t)](s) = \mathcal{L}[f](s-a)$ 。
- 微分性质： 若 $f^{(m)} \in C(\mathbb{R}_+)$ 且 $\lim_{t \rightarrow \infty} f^{(k)}(t) = 0$ ($k = 0, 1, 2, \dots, m-1$)，则

$$\mathcal{L}[f^{(m)}](s) = s^m \left(\mathcal{L}[f](s) - \sum_{k=0}^{m-1} \frac{1}{s^{k+1}} f^{(k)}(0) \right).$$

证明. 反复利用分部积分

$$\mathcal{L}[f^{(m)}] = \int_{\mathbb{R}} f^{(m)} e^{-st} dt = \int_{\mathbb{R}} e^{-st} df^{(m-1)} = f^{(m-1)}(0) + s \int_{\mathbb{R}} f^{(m-1)} e^{-st} dt = \dots$$

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Laplace 变换的卷积性质： $\mathcal{L}[f * g] = \mathcal{L}[f] \mathcal{L}[g]$ 。

证明. 按定义计算，并在计算中交换积分顺序

$$\begin{aligned} \mathcal{L}[f * g](s) &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(t-\tau) g(\tau) d\tau \right] e^{-st} dt \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(t-\tau) e^{-st} dt \right] g(\tau) d\tau \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(t-\tau) e^{-s(t-\tau)} dt \right] e^{-s\tau} g(\tau) d\tau \\ &= \mathcal{L}[f] \int_{\mathbb{R}} e^{-s\tau} g(\tau) d\tau = \mathcal{L}[f] \mathcal{L}[g]. \end{aligned}$$



Pierre-Simon, marquis de Laplace (1749-1827)

法国数学家，对工程、统计、物理、天文和哲学有重要贡献。

提出了 Laplace 方程，开创了 Laplace 变换。

被称为法国牛顿，被描述为拥有非凡的自然数学能力。

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Laplace 变换解偏微 (I): 有界杆上的热传导

$$\begin{cases} \partial_t u = \partial_x^2 u, & x \in (0, \ell), t > 0, \\ u(x, 0) = 1 + \sin \frac{\pi x}{\ell}, \\ u(0, t) = u(\ell, t) = 1 & t > 0. \end{cases}$$

对方程两边关于 t 做 Laplace 变换 (把 x 视为参数!!)。则

$$U(x, s) = \mathcal{L}[u(x, t)](s)$$

满足关于 x 的如下常微 (把 s 视为参数!!)

$$(\star) \begin{cases} sU(s, x) - u(x, 0) = U_{xx}(x, s), & t > 0, \\ U(0, s) = U(\ell, s) = \mathcal{L}[1](s) = \frac{1}{s}. \end{cases}$$

其解满足 $U(x, s) = A(s, x)e^{x\sqrt{s}} + B(s, x)e^{-x\sqrt{s}}$ 。用常数变易法得

$$U(x, s) = \frac{1}{s} + \frac{1}{s + \pi^2/\ell^2} \sin \frac{\pi x}{\ell}.$$

对上式做 Laplace 逆变换, 并记 $a = -\pi^2/\ell^2$, 我们有

$$u(x, t) = \mathcal{L}^{-1}[s^{-1}] + \sin \frac{\pi x}{\ell} \mathcal{L}^{-1}[(s - a)^{-1}].$$

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Laplace 变换解偏微 (II): 半无界杆上的热传导

$$\begin{cases} \partial_t u = \partial_x^2 u, & x \in (0, \infty), t > 0, \\ u(x, 0) = 0, & x \in (0, \infty), \\ u(0, t) = f(t), \lim_{x \rightarrow \infty} u(x, t) = 0, & t > 0. \end{cases}$$

对方程两边关于 t 做 Laplace 变换 (把 x 视为参数!!)。则

$$U(x, s) = \mathcal{L}[u(x, t)](s)$$

满足关于 x 的如下 ODE (把 s 视为参数!!)

$$(\star) \begin{cases} sU(s, x) = U_{xx}(x, s), & t > 0, \\ U(0, s) = \mathcal{L}[f](s), U(\infty, s) = 0. \end{cases}$$

其通解为 $U(x, s) = A(s)e^{-x\sqrt{s}} + B(s)e^{x\sqrt{s}}$ 。利用边值条件

$$U(x, s) = \mathcal{L}[f](s)e^{-x\sqrt{s}} = \mathcal{L}[f](s)\mathcal{L}\left[\frac{x}{2\sqrt{\pi}t^{3/2}}e^{-\frac{x^2}{4t}}\right].$$

对上式做 Laplace 逆变换, 并利用 Laplace 变换的卷积性质,

$$u(x, t) = \frac{x}{2\sqrt{\pi}} \int_0^t \frac{f(\tau)}{(t-\tau)^{3/2}} e^{-\frac{x^2}{4(t-\tau)}} d\tau.$$

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Laplace 变换解偏微 (III): 半无界弦振动

$$\begin{cases} \partial_t^2 u = \partial_x^2 u, & x \in (0, \infty), t > 0, \\ u(x, 0) = u_t(x, 0) = 0, & x \in (0, \infty), \\ u(0, t) = f(t), \lim_{x \rightarrow \infty} u(x, t) = 0, & t > 0. \end{cases}$$

对方程两边关于 t 做 Laplace 变换 (把 x 视为参数!!)。则

$$U(x, s) = \mathcal{L}[u(x, t)](s)$$

满足关于 x 的如下常微 (把 s 视为参数!!)

$$(\star) \begin{cases} s^2 U(s, x) = U_{xx}(x, s), & t > 0, \\ U(0, s) = \mathcal{L}[f](s), U(\infty, s) = 0. \end{cases}$$

其通解为 $U(x, s) = A(s)e^{-sx} + B(s)e^{sx}$ 。再由边值条件得

$$U(x, s) = \mathcal{L}[f](s)e^{-sx}.$$

对上式做 Laplace 逆变换, 并利用 Laplace 变换的卷积性质,

$$u(x, t) = \mathbf{1}_{\{t > x\}} f(t - x).$$

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Laplace 变换解偏微 (IV): 第一卦限的双曲方程

$$\begin{cases} \partial_{xy}^2 u = 1, & (x, y) \in \mathbb{R}_+^2, \\ u(x, 0) = 1, & x \in \mathbb{R}_+, \\ u(0, y) = y + 1, & y \in \mathbb{R}_+. \end{cases}$$

对方程两边关于 x 做 Laplace 变换 (把 y 视为参数!!)。则

$$U(s, y) = \mathcal{L}_x[u(x, y)](s)$$

满足关于 y 的如下 ODE (把 s 视为参数!!)

$$(\star) \begin{cases} \frac{d}{dy} [sU(s, y) - u(0, y)] = \frac{1}{s}, & y > 0, \\ U(s, 0) = \frac{1}{s}. \end{cases}$$

两边关于 y 积分, 得到 $(\star)$ 的解为

$$U(s, y) = \left(\frac{1}{s} + \frac{1}{s^2} \right) y + \frac{1}{s}$$

对上式做 Laplace 逆变换, 并利用 $\mathcal{L}_x[1] = \frac{1}{s}$ 和 $\mathcal{L}_x[x] = \frac{1}{s^2}$,

$$u(x, y) = (1 + x)y + 1.$$

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